

## Project Executable Files

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>Date</b>          | 1 October 2026                                    |
| <b>Team ID</b>       | SWTID-2026-7957                                   |
| <b>Project Name</b>  | EduGenie Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 3 Marks                                           |

### Project Executable Files

This document lists everything needed to run or deploy EduGenie: the submitted files, the folder structure, how to reach the live app, and how to run it locally.

#### Step 1: Submission Checklist

| S.No | Item to Submit                                                    | Submitted (Yes / No / NA) |
|------|-------------------------------------------------------------------|---------------------------|
| 1    | Complete source code (all files and folders)                      | Yes                       |
| 2    | README / Setup Guide (instructions to run the project)            | Yes                       |
| 3    | requirements.txt / package.json / dependency file                 | Yes                       |
| 4    | Database schema / seed files (if applicable)                      | NA                        |
| 5    | Environment configuration file (.env.example or similar)          | Yes                       |
| 6    | Deployed application URL (if hosted)                              | Yes                       |
| 7    | APK / Executable binary (if applicable for mobile / desktop apps) | NA                        |
| 8    | Dockerfile / Containerization config (if applicable)              | NA                        |
| 9    | Test files and test results                                       | Yes                       |
| 10   | Demo video or walkthrough (if required)                           | Yes                       |

#### Step 2: File / Folder Structure

```

EduGenie/
|-- main.py           FastAPI app and API endpoints
|-- gemini_client.py  shared Gemini helper (API key, model)
|-- explanation_module.py local LaMini-Flan-T5 explanations
|-- qna.py            question answering
|-- quiz_module.py    quiz generation and JSON validation
|-- summary_module.py summarization
|-- learning_path.py  learning recommendations
|-- templates/
|   |-- index.html    web page (HTML + JavaScript)
|-- static/
|   |-- style.css     styles
|-- requirements.txt  Python dependencies
|-- .env.example      environment template (copy to .env)
|-- .gitignore        keeps .env out of Git
|-- locustfile.py     load-test script
|-- README.md         setup and usage guide
  
```

### Step 3: Deployment / Access Details

| Field                       | Details                                                                                                                                                                             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hosted / Deployed URL       | <a href="https://edugenie-theta.vercel.app/">https://edugenie-theta.vercel.app/</a>                                                                                                 |
| Login Credentials (Demo)    | Not required. EduGenie is open access with no sign-up or login.                                                                                                                     |
| Platform / Hosting Provider | Vercel (web hosting); Google AI Studio (Gemini API key)                                                                                                                             |
| Repository Link             | <a href="https://github.com/mohdnishaal2007-rgb/EduGenie">https://github.com/mohdnishaal2007-rgb/EduGenie</a>                                                                       |
| Demo Video Link             | <a href="https://drive.google.com/file/d/1O0513n5u9dHq2CmGiOc-giZWM4zyDGI/view?usp=drivesdk">https://drive.google.com/file/d/1O0513n5u9dHq2CmGiOc-giZWM4zyDGI/view?usp=drivesdk</a> |

### Step 4: Run Instructions

1. Install Python 3.10 or newer and open the project folder in Visual Studio Code.
2. Create and activate a virtual environment: `python -m venv .venv`, then `.venv\Scripts\activate` (Windows) or `source .venv/bin/activate` (macOS / Linux). If PowerShell blocks activation, run `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass` first.
3. Install the dependencies: `pip install -r requirements.txt`
4. Copy `.env.example` to `.env`, then set `GEMINI_API_KEY` (from Google AI Studio) and `GEMINI_MODEL`.
5. Start the server: `uvicorn main:app --reload`
6. Open `http://127.0.0.1:8000`, pick a task from the dropdown, type your text and press the button.
7. Optional: run the load test with `locust -f locustfile.py --host http://127.0.0.1:8000`.

### Step 5: Known Issues / Limitations

| S.No | Known Issue / Limitation                                                                                   | Workaround / Status                                                                       |
|------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1    | The first concept explanation downloads the local model (about 3 GB), so it is slow once.                  | Run one warm-up explanation beforehand; if the model cannot load, Gemini answers instead. |
| 2    | There is no login, saved history or progress tracking.                                                     | Planned as a future enhancement (see Scalability & Future Plan).                          |
| 3    | AI answers can occasionally be inaccurate or incomplete.                                                   | Learners should check important facts against textbooks or teachers.                      |
| 4    | Gemini features need internet and are limited by the API key's quota; Google also retires old model names. | Change <code>GEMINI_MODEL</code> in the environment settings; no code change is needed.   |